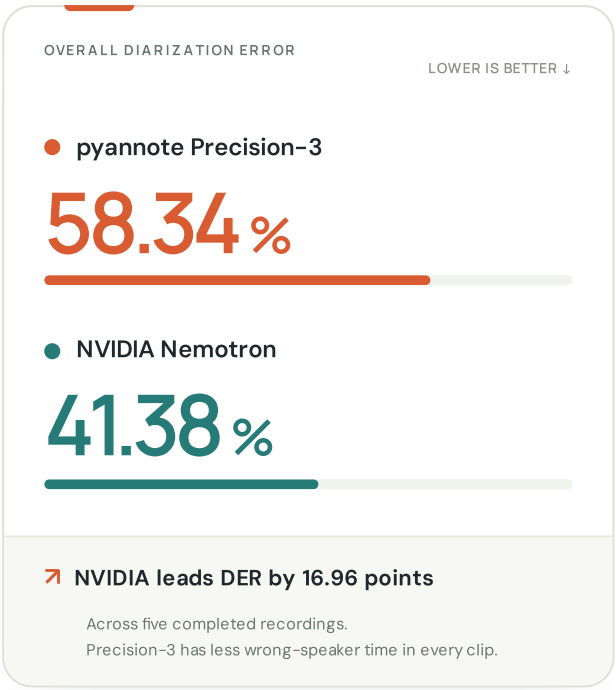
checksub.

SPEAKER DIARIZATION / UPDATED 2 NOV 2026

Speaker detection, measured.

pyannote Precision-3 and NVIDIA Nemotron 3 Diarization, tested on five English recordings against shared, human-reviewed subtitle references.



THE TEST SET

Same audio, both detectors

05	40m 35s	5–11	3 / 5
Completed English recordings	Total audio evaluated	Reference speakers per clip	Clips favor NVIDIA on DER

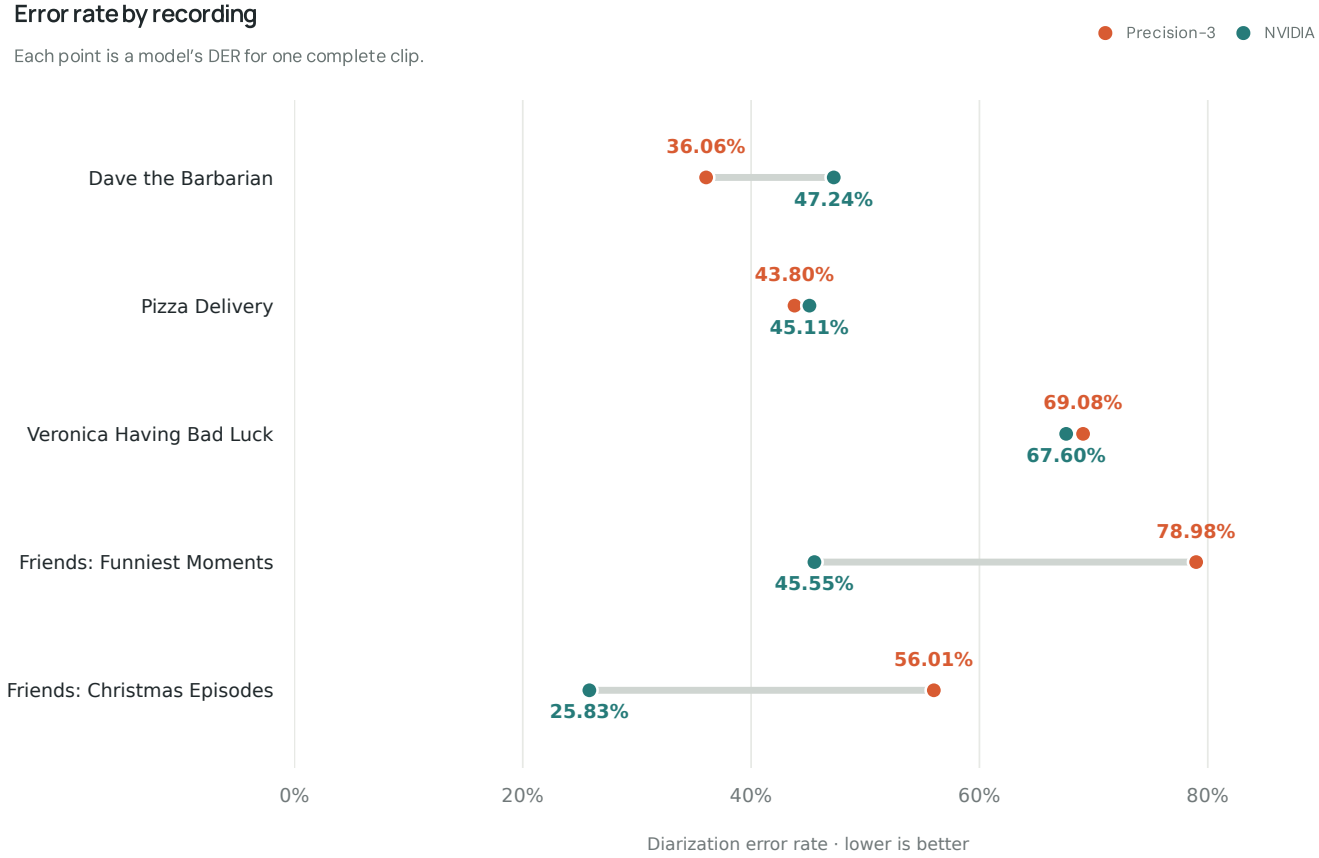
01 THE RESULTS

NVIDIA leads on DER. Precision-3 on speaker confusion.

NVIDIA has lower weighted DER across the five clips. Precision-3 has less wrong-speaker time in every clip. The difference comes mainly from speech detected outside the subtitle reference.

Error rate by recording

Each point is a model's DER for one complete clip.



0-second collar · overlapping speech included

DER is an error rate, not a percentage accuracy score.

WITHIN THE EIGHT-SPEAKER LIMIT

The three clips within the limit.

Pizza, Veronica and Friends: Funniest Moments have at most eight reference speakers. NVIDIA also leads DER on this subset, by **17.12 percentage points**. Subtitle timing limits still apply.

Precision-3

67.71%

Three recordings · weighted DER

NVIDIA

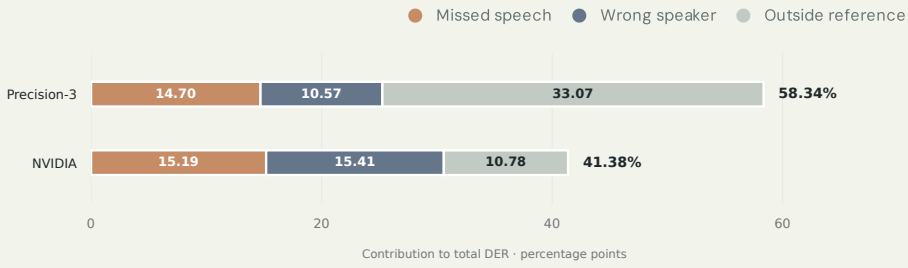
50.59%

Three recordings · weighted DER

LOOKING INSIDE THE SCORE

What went wrong?

Precision-3 totals **139.68 seconds** of wrong-speaker time; NVIDIA totals **203.66 seconds**. NVIDIA's lower DER comes mainly from fewer detections outside the reference. Those intervals can contain real speech omitted from subtitles and need listening review.



How we calculate DER ↗

Counts can be too low, too high or exactly right while individual assignments remain wrong. The brief ninth

Every recording tells a different story.

character in Friends Christmas contributes only 2.12 seconds of reference speech.

01

4:23 · 11 reference voices

Dave the Barbarian

Singing, music and a large cast

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. This may favor agreement with the Checksub baseline represented here by Precision-3. Cue times came from Checksub's transcription/alignment pipeline and were not independently annotated.

Precision-3	36.06%
NVIDIA	47.24%
Speaker labels Reference / Precision-3 / NVIDIA	11 / 9 / 5

Precision-3 lower error

02

4:04 · 5 reference voices

Pizza Delivery

Rapid dialogue and background music

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. This may favor agreement with the Checksub baseline represented here by Precision-3. Cue times came from Checksub's transcription/alignment pipeline and were not independently annotated.

Precision-3	43.80%
NVIDIA	45.11%
Speaker labels Reference / Precision-3 / NVIDIA	5 / 4 / 3

Precision-3 lower error

03

3:59 · 8 reference voices

Veronica Having Bad Luck

Short turns and similar character voices

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. This may favor agreement with the Checksub baseline represented here by Precision-3. Cue times came from Checksub's transcription/alignment pipeline and were not independently annotated.

Precision-3	69.08%
NVIDIA	67.60%
Speaker labels Reference / Precision-3 / NVIDIA	8 / 5 / 5

NVIDIA lower error

04

14:01 · 8 reference voices

Friends: Funniest Moments

Live-action dialogue, laughter and long subtitle gaps

Reference caveat

Reviewed from Checksub's original subtitle export and speaker assignments. This may favor agreement with the Checksub baseline represented here by Precision-3. Cue times came from Checksub's transcription/alignment pipeline and were not independently annotated.

Precision-3	78.98%
NVIDIA	45.55%
Speaker labels Reference / Precision-3 / NVIDIA	8 / 9 / 8

NVIDIA lower error

05

14:08 · 9 reference voices

Friends: Christmas Episodes

Short interjections, laughter and a brief ninth speaker

Reference caveat

Reviewed from an SRT prepared with NVIDIA speaker labels and existing Checksub cue times. This may favor NVIDIA's measured agreement, including how well its output aligns with the reference. The cue times were inherited from Checksub and were not independently annotated.

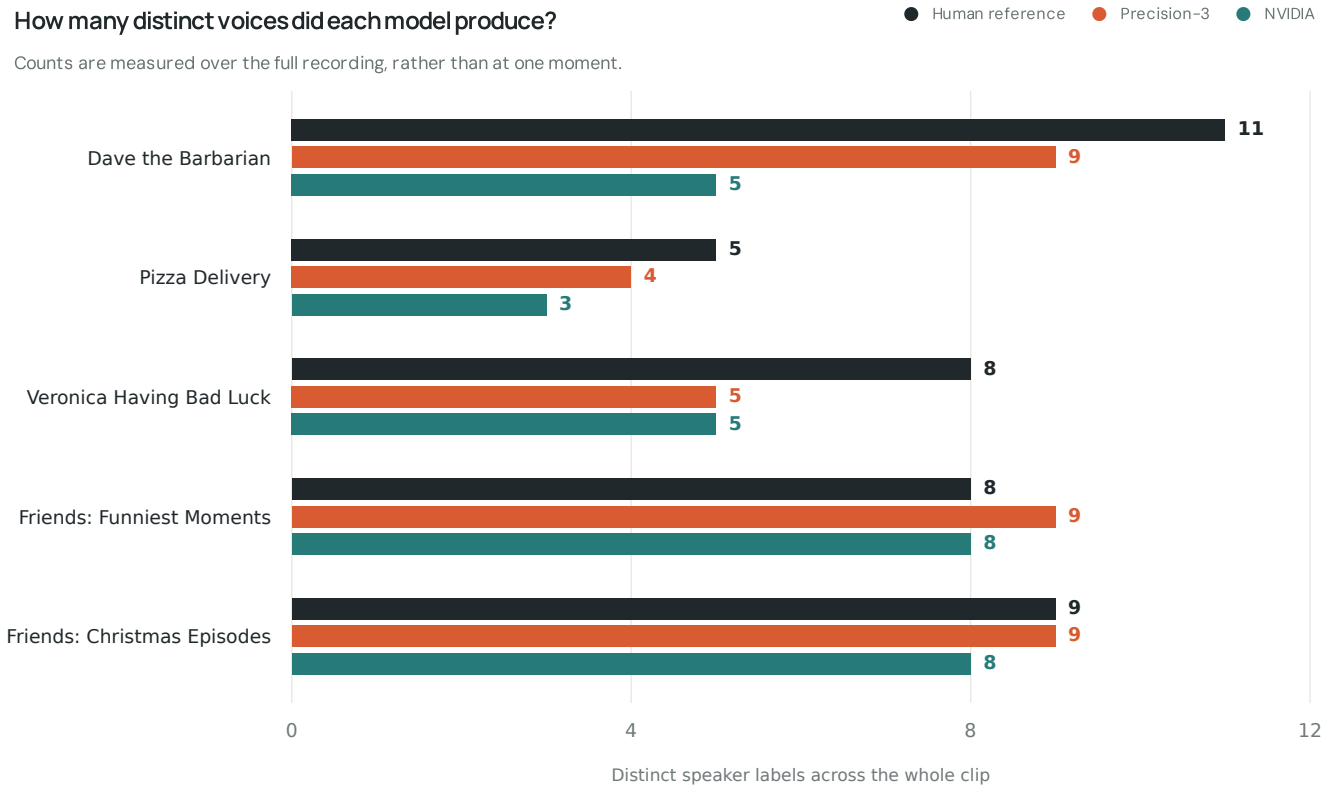
Precision-3	56.01%
NVIDIA	25.83%
Speaker labels Reference / Precision-3 / NVIDIA	9 / 9 / 8

NVIDIA lower error

A material limit on this comparison: Dave has 11 reference speakers and Friends Christmas has nine. NVIDIA supports at most eight speakers per session. Both recordings exceed its capacity; the separate three-recording subset stays within the limit.

How many distinct voices did each model produce?

Counts are measured over the full recording, rather than at one moment.



03 AN INTERACTIVE CLOSER LOOK

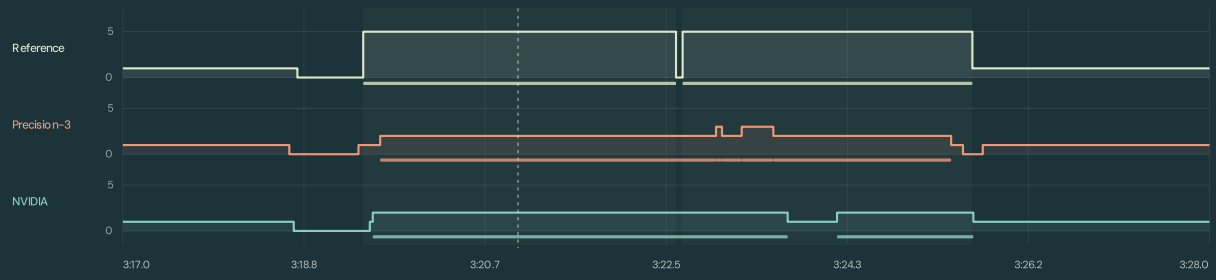
Inside the overlap.

Both models flag simultaneous voices. Explore exactly when they do, and how many speaker labels are active at the same moment.

DAVE THE BARBARIAN / FIVE-VOICE CHORUS

Five voices in the reference. Fewer in the predictions.

Human reference · Precision-3 · NVIDIA · Height = simultaneous speaker labels



3:21.00 Reference 5 Precision-3 2 NVIDIA 2

Annotated overlap

6.10s

Two chorus passages, with five reference characters singing together.

Precision-3 · within the chorus

5.71s

93.7% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

NVIDIA · within the chorus

5.50s

90.2% of annotated overlap marked with ≥2 labels. This does not verify speaker identities.

When was overlap detected?

3 intervals

TIME WINDOW	REFERENCE PEAK	PRECISION-3 PEAK	NVIDIA PEAK	REFERENCE STATUS
1:08.96–1:08.97 ↗	1	1	2	No overlap annotation
3:19.43–3:22.60 ↗	5	2	2	Annotated overlap
3:22.67–3:25.60 ↗	5	3	2	Annotated overlap

The two annotated intervals are 3:19.43–3:22.60 and 3:22.67–3:25.60. Candy, Mrs. Bogmelon, Oswidge, Faffy and Fang each have a reference row. Model-only flags outside those intervals are not independently verified. Event rows are clickable and show peak simultaneous counts inside each interval.

Detecting overlap and recovering every voice are separate tasks. In the annotated chorus, Precision-3 reaches three simultaneous labels; NVIDIA reaches two. The reference contains five. Those counts do not verify which characters were assigned correctly.

04 HOW TO READ THESE NUMBERS

These are our measured results, not the vendors' published benchmark scores. They describe this test set

A useful pilot. A specific set of limits.

and these references.

THE METRIC

Diarization error rate

$$\frac{\text{Missed speech + extra speech + wrong-speaker time}}{\text{Reference speaker time}}$$

Multiplied by 100. **Lower is better.** DER can exceed 100%; subtracting it from 100 does not give a reliable "accuracy" score.

Scores are weighted by reference speaker time. Overlapping reference voices each contribute speaker time.

01 Identical audio, isolated detection

Both models received the same 16 kHz mono WAV for each clip. This compares Precision-3 with Nemotron 3 Diarization, not the full Nemotron 3.5 Transcribe pipeline.

02 Model-assisted labels, subtitle-based times

The first four references began with Checksub's original subtitle exports and speaker assignments, which may favor the Checksub/Precision-3 baseline. Friends Christmas began with NVIDIA speaker labels, which may favor NVIDIA's measured agreement. All five use Checksub transcription/alignment cue times; neither diarization model independently supplied those timings. Human review does not make this a blind reference. Only Dave's chorus has explicit overlap annotation.

03 Strict, consistent scoring

Zero boundary collar; overlap included; full-clip evaluation. Speaker IDs are matched to names globally and one-to-one, so splitting or merging identities is penalized.

04 A small, specialized sample

Three animation clips and two Friends compilations, all in English, with music, laughter and short turns. Dave and Friends Christmas exceed NVIDIA's eight-speaker capacity. Customer recordings and other languages need separate evaluation.

05 Speed and cost need a separate test

Precision-3 used a hosted API; NVIDIA used local GPU inference. Queueing, upload and model load have different scopes. This report makes no apples-to-apples speed or cost claim.

05 CONCLUSION

Keep Precision-3 as the baseline, for now.

For assigning subtitles to the correct person, keep Precision-3 as the provisional default. It has **less wrong-speaker time in all five clips**. NVIDIA leads total DER, largely through fewer detections outside the subtitle reference.

NVIDIA remains a promising local candidate. The four original references may favor the Checksub baseline; Christmas may favor NVIDIA. Review speech boundaries and identities independently before a replacement decision. Neither system recovers all five annotated chorus voices.

● PROVISIONAL RECOMMENDATION

BEFORE A REPLACEMENT DECISION

1. **Tighten the reference**

   Annotate actual speech boundaries, identities and overlaps independently of subtitle display timing and either model's predictions. Check the intervals currently counted as extra speech.

2. **Broaden and match the test set**

   Add customer-style interviews in our working languages; compare recordings within the eight-speaker limit, with larger casts assessed separately.

3. **Measure the complete trade-off**

   Compare correction time, deployment cost and latency under consistent conditions, then inspect the final Checksub editor output.

checksub.

Speaker detection evaluation · 2 November 2026

Five completed recordings · exploratory study